

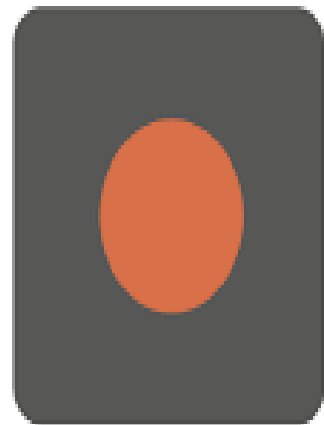


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Introduction to Statistical Methods(ISM)

Course No: AIML ZC418
Course Title: ISM
WEBINAR 4 : 23-04-2026

IMP Note to Self



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Problems based on the following topics will be covered in this webinar:

- Statistical Inference
- Maximum Likelihood
- ANOVA (one way and two ways)
- Corelation



Statistical Inference

Three forms of statistical inference

- Point estimation
- Interval estimation
- Hypothesis testing



Point Estimate

- A **point estimate** is a statistic taken from a sample that is used to estimate a population parameter.
- A point estimate is only as good as the representativeness of its sample.
- If other random samples are taken from the population, the point estimates derived from those samples are likely to vary.

Terminology and Notations

**Statistical
measure**

**Sample
estimates**

**Population
parameter**

Mean

$\bar{x}$

μ

Variance

s^2

σ^2

Proportion

$\bar{p}$

P

Correlation

r

ρ

Confidence interval for Mean

A $100(1 - \alpha)\%$ confidence interval for the mean μ of a normal population when the value of σ is known is given by

$$\left(\bar{x} - z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} \right) \quad (7.5)$$

or, equivalently, by $\bar{x} \pm z_{\alpha/2} \cdot \sigma / \sqrt{n}$.



If $\bar{x}$ is used as an estimate of μ , we can be $100(1 - \alpha)\%$ confident that the error will not exceed a specified amount e when the sample size is

$$n = \left(\frac{z_{\alpha/2} \sigma}{e} \right)^2.$$

Confidence interval for Mean

If n is sufficiently large, the standardized variable

$$Z = \frac{\bar{X} - \mu}{S/\sqrt{n}}$$

has approximately a standard normal distribution. This implies that

$$\bar{x} \pm z_{\alpha/2} \cdot \frac{s}{\sqrt{n}} \quad (7.8)$$

is a **large-sample confidence interval for μ** with confidence level approximately $100(1 - \alpha)\%$. This formula is valid regardless of the shape of the population distribution.

Problem 1:

Assume that the helium porosity (in percentage) of coal samples taken from any particular seam is normally distributed with true standard deviation .75.

Compute a 98% Confidence Interval for the true average porosity of a certain seam if the average porosity for 16 specimens from the seam was 4.56.

Solution:

$$n = 16; \bar{X} = 4.56; \sigma = 0.75; 1 - \alpha = 0.98 \Rightarrow \alpha = 0.02; Z_{\frac{\alpha}{2}} = 2.33$$

Confidence interval for mean is $\bar{X} - Z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}} < \mu < \bar{X} + Z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}$

$$4.56 - 2.33 \times \frac{0.75}{\sqrt{16}} < \mu < 4.56 + 2.33 \times \frac{0.75}{\sqrt{16}}$$

$$4.56 - 0.437 < \mu < 4.56 + 0.437$$

$$4.123 < \mu < 4.997$$

Problem 2:

A random sample of 300 Citi Bank VISA cardholder accounts indicated a sample mean debt of \$1,220 with a sample standard deviation of \$840. Construct a 95 percent confidence interval estimate of the average debt of all cardholders.

Solution:

$$n=300 ; \bar{X} = 1220 ; s = 840 ; 1 - \alpha = 0.95 \Rightarrow \alpha = 0.05 ; Z_{\frac{\alpha}{2}} = 1.96$$

Confidence interval for mean is

$$\bar{X} - Z_{\frac{\alpha}{2}} \frac{s}{\sqrt{n}} < \mu < \bar{X} + Z_{\alpha/2} \frac{s}{\sqrt{n}}$$

$$1220 - 1.96 \times \frac{840}{\sqrt{300}} < \mu < 1220 + 1.96 \times \frac{840}{\sqrt{300}}$$

$$1220 - 95.05 < \mu < 1220 + 95.05$$

$$1124.95 < \mu < 1315.05$$

Problem 3

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A random sample of 400 items is found to have a mean of 82 and standard deviation of 18. Find 99% confidence interval for the mean of the population from which the sample is drawn.

Solution:

$$n = 400; \bar{X} = 82; s = 18; 1 - \alpha = 0.99 \Rightarrow \alpha = 0.01; Z_{\frac{\alpha}{2}} = 2.575$$

Confidence interval for mean is $\bar{X} - Z_{\frac{\alpha}{2}} \frac{s}{\sqrt{n}} < \mu < \bar{X} + Z_{\alpha/2} \frac{s}{\sqrt{n}}$

$$82 - 2.575 \times \frac{18}{\sqrt{400}} < \mu < 82 + 2.575 \times \frac{18}{\sqrt{400}}$$

$$82 - 2.3175 < \mu < 82 + 2.3175$$

$$79.6825 < \mu < 84.3175$$

Problem 4:

The article “Evaluating Tunnel Kiln Performance” (*Amer. Ceramic Soc. Bull.*, Aug. 1997: 59–63) gave the following summary information for fracture strengths (MPa) of $n=169$; ceramic bars fired in a particular kiln: $\bar{X} = 89.10$; $s = 3.73$

- a. Calculate a (two-sided) confidence interval for true average fracture strength using a confidence level of 95%.
- b. Suppose the investigators had believed *a priori* that the population standard deviation was about 4 MPa. Based on this supposition, how large a sample would have been required to estimate μ to within .5 MPa with 95% confidence?

Solution:

a) $n = 169$; $\bar{X} = 89.10$; $s = 3.73$; $1 - \alpha = 0.95 \Rightarrow \alpha = 0.05$; $Z_{\frac{\alpha}{2}} = 1.96$

$$\bar{X} - Z_{\frac{\alpha}{2}} \frac{s}{\sqrt{n}} < \mu < \bar{X} + Z_{\alpha/2} \frac{s}{\sqrt{n}} \qquad 89.10 - 1.96 \times \frac{3.73}{\sqrt{169}} < \mu < 89.10 + 1.96 \times \frac{3.73}{\sqrt{169}}$$

$$89.10 - 0.5624 < \mu < 89.10 + 0.5624$$

$$88.54 < \mu < 89.66$$

Solution contd.,

$$b) \ n = ? ; \bar{X} = 89.10 ; s = 4 ; 1 - \alpha = 0.95 \Rightarrow \alpha = 0.05 ; Z_{\frac{\alpha}{2}} = 1.96 ;$$

margin of error , $E = 0.5$

$$\begin{aligned} E &= Z_{\alpha/2} \times \frac{s}{\sqrt{n}} \\ \Rightarrow 0.5 &= 1.96 \times \frac{4}{\sqrt{n}} \\ \Rightarrow n &= \left(1.96 \times \frac{4}{0.5} \right)^2 = 245.86 \end{aligned}$$

Required sample size is 246



Maximum Likelihood Estimation (MLE)



Method of Maximum Likelihood Estimation is the best and most popular one among all methods to obtain an almost good or best estimator for a population parameter.

It is a method of obtaining an estimator which most (maximum) likely estimates the true value of the parameter i.e., finding an estimator that can give most likely nearer value for the unknown true value of parameter.

The corresponding estimator is called maximum likelihood estimator (MLE).

MLE for a Binomial distribution parameter



Suppose we wish to find the MLE of θ for a Binomial distribution with k successes in n trials then,

$$p_k(k, \theta) = nC_k \theta^k (1 - \theta)^{n-k}$$

$$\log p_k(k, \theta) = \log(nC_k) + k \log(\theta) + (n - k) \log((1 - \theta))$$

$$\frac{\partial \log p_k(k, \theta)}{\partial \theta} = 0 \Rightarrow 0 + \frac{k}{\theta} - \frac{n - k}{1 - \theta} = 0$$

$$k - k\theta = n\theta - k\theta \Rightarrow \theta = \frac{k}{n}$$

MLE for Poisson Distribution Parameter



- Suppose we have data generated from a Poisson distribution. We want to estimate the parameter of the distribution
- The probability of observing a particular random variable is $P(X; \mu) = \frac{e^{-\mu} \mu^X}{X!}$
- Joint likelihood by multiplying the individual probabilities together

$$P(X_1, X_2, \dots, X_n; \mu) = \frac{e^{-\mu} \mu^{X_1}}{X_1!} \times \frac{e^{-\mu} \mu^{X_2}}{X_2!} \times \dots \times \frac{e^{-\mu} \mu^{X_n}}{X_n!}$$

$$L(\mu; \mathbf{X}) = \prod_i e^{-\mu} \mu^{X_i}$$

$$L(\mu; \mathbf{X}) = e^{-n\mu} \mu^{n\bar{X}}$$

MLE for Poisson Distribution Parameter



- Note in the likelihood function the factorials have disappeared.
- This is because they provide a constant that does not influence the relative likelihood of different values of the parameter
- It is usual to work with the **log likelihood** rather than the likelihood.
- Note that maximising the log likelihood is equivalent to maximising the likelihood.

$$L(\mu; \mathbf{X}) = e^{-n\mu} \mu^{n\bar{X}}$$

Take the natural log of the likelihood function

$$\ell(\mu; \mathbf{X}) = -n\mu + n\bar{X} \log \mu$$

Find where the derivative of the log likelihood is zero

$$\frac{d\ell}{d\mu} = -n + \frac{n\bar{X}}{\mu}$$

Note that here the MLE is the same as the moment estimator

$$\hat{\mu} = \bar{X}$$

MLEs for Normal Distribution Parameters

- Let $X_1, \dots, X_n$ be a random sample from a normal distribution.
- The likelihood function is

$$\begin{aligned} f(x_1, \dots, x_n; \mu, \sigma^2) &= \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x_1 - \mu)^2 / (2\sigma^2)} \cdot \dots \cdot \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x_n - \mu)^2 / (2\sigma^2)} \\ &= \left(\frac{1}{2\pi\sigma^2} \right)^{n/2} e^{-\sum (x_i - \mu)^2 / (2\sigma^2)} \end{aligned}$$

- so

$$\ln[f(x_1, \dots, x_n; \mu, \sigma^2)] = -\frac{n}{2} \ln(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum (x_i - \mu)^2$$

MLEs for Normal Distribution Parameters



- To find the maximizing values of μ and σ^2 , we must take the partial derivatives of $\ln(f)$ with respect to μ and σ^2 , equate them to zero, and solve the resulting two equations.
- Omitting the details, the resulting MLE's are

$$\hat{\mu} = \bar{X} \quad \hat{\sigma}^2 = \frac{\sum (X_i - \bar{X})^2}{n}$$

- The MLE of σ^2 is not the unbiased estimator, so two different principles of estimation (unbiasedness and maximum likelihood) yield two different estimators

MLEs

- If θ denotes the parameter to be estimated, then estimators will be denoted by $\hat{\theta}$

- For Bernoulli
$$\hat{p} = \frac{\sum_{i=1}^n x_i}{n} = \bar{X}$$
 For Exponential $\hat{\lambda} = 1/\bar{X}$

- For Poisson's
$$\hat{\lambda} = \frac{\sum_{i=1}^n x_i}{n}$$

- For Normal
$$\hat{\mu} = \frac{\sum_{i=1}^n x_i}{n}$$

$$\hat{\sigma} = \sqrt{\frac{\sum_{i=1}^n (x_i - \hat{\mu})^2}{n}}$$

Problem 1:

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The number of defective hard drives produced daily by a production line can be modeled as Poisson's distribution. The counts for ten days are 7 3 1 2 4 1 2 3 1 2. Obtain max likelihood estimate of probability of 0 or 1 defectives on one day.

Solution

From the maximum likelihood estimate of λ is $\hat{\lambda} = \bar{x} = 26/10 = 2.6$. Consequently, by the invariance property, the maximum likelihood estimate of

$$P(X = 0 \text{ or } 1) = e^{-\lambda} + \frac{\lambda e^{-\lambda}}{1!}$$

is

$$e^{-\hat{\lambda}} + \frac{\hat{\lambda} e^{-\hat{\lambda}}}{1!} = e^{-2.6} + \frac{2.6 \cdot e^{-2.6}}{1!} = 0.267$$

There will 1 or fewer defectives on just over one-quarter of the days.

Problem 2:

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Consider a sample 0,1,0,0,1,0 from a binomial distribution, with the form $P[X=0]=(1-p)$, $P[X=1]=p$. Find the maximum likelihood estimate of p .

Solution :

$$\begin{aligned} L(p) &= P[X=0] P[X=1] P[X=0] P[X=0] P[X=1] P[X=0] \\ &= (1-p) p (1-p) (1-p) p (1-p) \\ &= (1-p)^4 p^2. \end{aligned}$$

$$\text{Log } L(p) = \log[(1-p)^4 p^2] = \log[(1-p)^4] + \log[(p^2)] = 4\log(1-p) + 2\log p$$

$$\frac{\partial \text{Log} L(p)}{\partial p} = 0 \quad \text{means,} \quad \frac{-4}{1-p} + \frac{2}{p} = 0 \Rightarrow \frac{-4p+2-2p}{p(1-p)} = 0 \Rightarrow p = 1/3$$

That is , there is 1/3 chance to observe this sample if we believe the population to be Binomial distributed .

One-way Analysis of Variance

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Treatment-1	Treatment-2	Treatment-3	.	.	Treatment-k
X_{11}	X_{21}	X_{31}	.	.	X_{k1}
X_{12}	X_{22}	X_{32}	.	.	X_{k2}
X_{13}	X_{23}	X_{33}	.	.	X_{k3}
.	.	.	.	.	.
.	.	.	.	.	.
.	.	.	.	.	.

$$n_1 + n_2 + n_3 + \dots + n_k = n$$

$$C_1 + C_2 + C_3 + \dots + C_k = G$$

**The hypotheses
to be tested are**

$$H_0 : \mu_1 = \mu_2 = \dots = \mu_k$$

vs

$$H_1 : \mu_1 \neq \mu_2 \neq \dots \neq \mu_k$$

- The estimate of σ^2 based on the variation of the sample means is called the mean square due to treatments and is denoted by MSTR

$$\text{MSTR} = \frac{\sum_{j=1}^k n_j (\bar{x}_j - \bar{\bar{x}})^2}{k - 1}$$

Denominator is the degrees of freedom associated with SSTR

Numerator is called the sum of squares due to treatments (SSTR)

- The estimate of σ^2 based on the variation of the sample observations **within each sample** is called the mean square error and is denoted by MSE

$$\text{MSE} = \frac{\sum_{j=1}^k (n_j - 1)s_j^2}{n - k}$$

Denominator is the degrees of freedom associated with SSE

Numerator is called the sum of squares due to error (SSE)

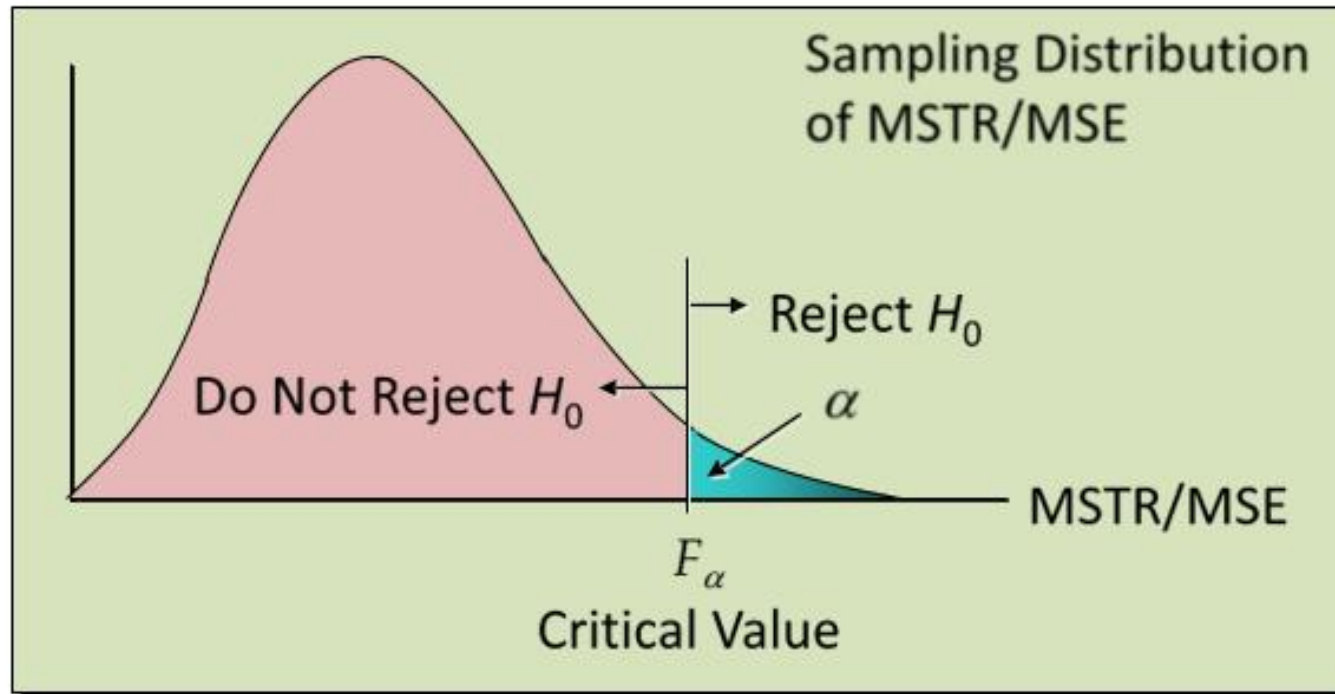
F-Test (One way ANOVA)

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- If the null hypothesis is true and the ANOVA assumptions are valid, the sampling distribution of $MSTR/MSE$ is an F distribution with $MSTR$ d.f equal to $k - 1$ and MSE d.f. equal to $n - k$.



Solution Method-2: using Correction Factor

Step 1: Correction Factor (CF) = G^2 / n

Step 2: $SST = \sum \sum x_{i,j}^2 - CF$

Step 3: $SSTR = \frac{\sum c_i^2}{n_i} - CF$

Step 4: SSE

Step 5: ANOVA Table

Grand total , G =150

One way ANOVA Table

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Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F
Treatments	SSTR	$k - 1$	$MSTR = \frac{SSTR}{k-1}$	$\frac{MSTR}{MSE}$
Error	SSE	$n - k$	$MSE = \frac{SSE}{n-k}$	
Total	SST	$n - 1$		

SST is partitioned into SSTR and SSE.

SST's degrees of freedom (d.f.) are partitioned into SSTR's d.f. and SSE's d.f.

$$SST = SSTR + SSE$$

One way ANOVA Example

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Problem 1: Agricultural Yields

A researcher wants to determine if three different types of fertilizer (A, B, and C) result in different average crop yields. Five plots of land are assigned to each fertilizer type, and the yield (in bushels) is recorded for each.

Fertilizer A	Fertilizer B	Fertilizer C
20	25	18
22	28	17
19	22	20
21	26	16
23	29	19

Questions:

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1. **State the Hypotheses:** Define H_0 (Null) and H_1 (Alternative).
2. **Calculate the Variance:** Determine the "Sum of Squares Between" (SSB) and "Sum of Squares Within" (SSW).
3. **The F-Statistic:** Using a significance level of $\alpha = 0.05$, determine if there is a statistically significant difference in yields.

Scenario: Do Fertilizers A, B, and C result in different average crop yields?

Null Hypothesis (H0): $\mu_1 = \mu_2 = \mu_3$ (All fertilizers have the same effect)

Alt Hypothesis (H1): At least one mean is different.

Fertilizer A	Fertilizer B	Fertilizer C
20, 22, 19, 21, 23	25, 28, 22, 26, 29	18, 17, 20, 16, 19
Mean: 21.0	Mean: 26.0	Mean: 18.0

Grand Mean ($\bar{\bar{X}}$): 21.67

ANOVA Table

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Source	SS	df	MS	F
Between Groups	163.33	2	81.67	19.600
Within Groups (Error)	50.00	12	4.17	
Total	213.33	14	-	-

Conclusion: With F-crit (2, 12) at $\alpha=0.05 \approx 3.89$. Since $19.600 > 3.89$, we **Reject H_0** .
Fertilizer type significantly affects yield.

Problem2: Battery Life Expectancy

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An electronics company is testing the longevity of three different brands of AAA batteries (**Duramax**, **Energize**, and **Volt+**) used in high-drain devices. They track how many hours each battery lasts before failing.

- **Group 1 (Duramax):** 12, 15, 14, 11, 13
- **Group 2 (Energize):** 18, 16, 17, 19, 20
- **Group 3 (Volt+):** 13, 12, 11, 14, 12

Questions:

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1. **State the Hypotheses:** Define H_0 (Null) and H_1 (Alternative).
2. **Calculate the Variance:** Determine the "Sum of Squares Between" (SSB) and "Sum of Squares Within" (SSW).
3. **The F-Statistic:** Using a significance level of $\alpha = 0.05$, determine if there is a statistically significant difference in yields.

Problem -2 Battery Life

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Scenario: Do Duramax, Energize, and Volt+ differ in longevity?

Brand	Data (Hours)	Mean ($\bar{x}$)
Duramax	12, 15, 14, 11, 13	13.0
Energize	18, 16, 17, 19, 20	18.0
Volt+	13, 12, 11, 14, 12	12.4

Grand Mean: 14.47

ANOVA Table

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Source	SS	df	MS	F
Between Groups	94.53	2	47.27	22.508
Within Groups (Error)	25.20	12	2.10	

Conclusion: F-crit (2, 12) at $\alpha=0.05 \approx 3.89$. Since $22.508 > 3.89$, we **Reject H_0** . There is a highly significant difference in battery life between the brands.

Treatments

Blocks

	1	2	...	k	Total
1	x_{11}	x_{12}	...	x_{1k}	$x_{.1}$
2	x_{21}	x_{22}	...	x_{2k}	$x_{.2}$
3	x_{31}	x_{32}	...	x_{3k}	$x_{.3}$
...	...	...	...	...	...
m	x_{m1}	x_{m2}	...	x_{mk}	$x_{.m}$
Total	$x_{1.}$	$x_{2.}$	...	$x_{k.}$	G

Step 1: In two-way ANOVA we have two pairs of hypotheses, one for treatments and one for the blocks.

Null Hypotheses

H_01 : There is no significant difference among the population means of different Treatments(groups)

H_02 : There is no significant difference among the population means of different Blocks

Alternative Hypotheses

H_11 : At least one pair of treatment means differs significantly

H_12 : At least one pair of block means differs significantly

Step 2: Data is presented in a rectangular table form as described in one-way ANOVA.

Step 3: Level of significance α .

Step 4: Test Statistic

For treatments $F = \text{MSTR} / \text{MSE}$

For blocks $F = \text{MSBL} / \text{MSE}$

Steps involved in two-way ANOVA

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To find the test statistic we have to find the following intermediate values.

i) **Correction Factor:** $C.F = \frac{G^2}{n}$ where $G = \sum_{j=1}^m \sum_{i=1}^k x_j$

ii) **Total Sum of Squares:** $SST = \sum_{i=1}^k \sum_{j=1}^m x_{ij}^2 - C.F$

iii) **Sum of Squares between Treatments:** $SSTR = \sum_{i=1}^k \frac{x_i^2}{m} - C.F$

iv) **Sum of squares between blocks:** $SSBL = \sum_{j=1}^m \frac{x_j^2}{k} - C.F$

v) **Sum of Squares due to Error:** $SSE = SST - SSTR - SSBL$

Problem 3: Retail Sales Performance

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A retail chain wants to see how **Store Layout** and **Background Music** affect the average daily sales (in thousands of dollars). They test two types of layouts and three types of music across several stores.

Store Layout	No Music	Classical	Pop
Grid Layout	10, 12	14, 15	18, 20
Free-Flow Layout	11, 13	19, 21	22, 24

Questions

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1. **Identify the Factors:** What are the two independent variables?
2. **State the Three Null Hypotheses:** * H_{01} : Store layout has no effect on sales.
 - H_{02} : Music type has no effect on sales.
 - H_{03} : There is no interaction between layout and music.
3. **Analysis:** If the interaction effect is significant, what does that tell the manager about how music works in different layouts?

Null Hypotheses:

$H_{01}: \mu_{Grid} = \mu_{FreeFlow}$ Store layout has no effect on sales

$H_{02}: \mu_{NoMusic} = \mu_{Classical} = \mu_{Pop}$

Interaction Effect

$H_{03}: \text{No interaction between Layout and Music}$

Problem4: Battery Life

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An engineer is testing the life of a new battery (in hours) based on two factors: **Battery Material** (Lead, Lithium, Nickel) and **Operating Temperature** (Low, Medium, High).

Material	Low Temp (10°C)	Med Temp (30°C)	High Temp (50°C)
Lead	100, 105	80, 85	50, 55
Lithium	150, 155	140, 145	120, 125
Nickel	120, 122	110, 115	90, 95

Problem 3:

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- An experiment was performed to determine the effect of four different chemicals on the strength of a fabric.
- These chemicals are used as part of the permanent press finishing process.
- Five fabric samples were selected, and a randomized complete block design was run by testing each chemical type once in random order on each fabric sample.
- The data are shown in Table.
- Test for differences in means using an ANOVA with $\alpha = 0.05$.

Chemical Type	Fabric Sample				
	1	2	3	4	5
1	1.3	1.6	0.5	1.2	1.1
2	2.2	2.4	0.4	2.0	1.8
3	1.8	1.7	0.6	1.5	1.3
4	3.9	4.4	2.0	4.1	3.4

In general for computation purpose the following formula can be used

$$r = \frac{n \sum XY - (\sum X)(\sum Y)}{\sqrt{n \sum X^2 - (\sum X)^2} \sqrt{n \sum Y^2 - (\sum Y)^2}}$$

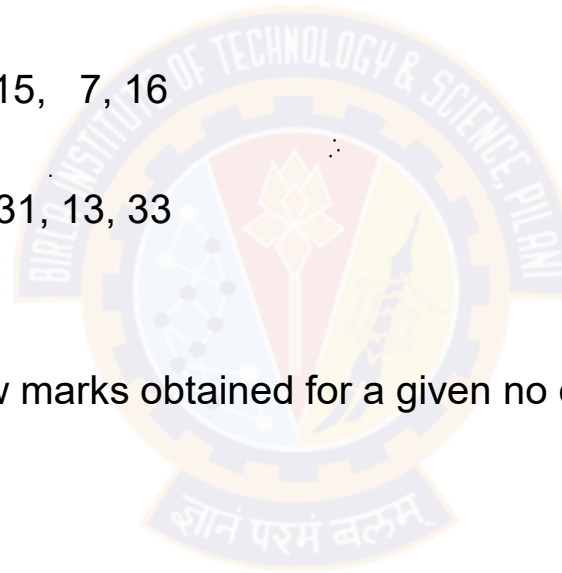
- Calculate the Karl Pearson's coefficient for the following data and draw your conclusion:

No of Study Hours: 10, 12, 13, 9, 11, 14, 8, 15, 7, 16

Marks Obtained : 19, 25, 26, 17, 23, 29, 15, 31, 13, 33

For the same data draw the scatter plot.

Also find the linear regression that predicts the marks obtained for a given no of study hours.



Problem 5

Consider the following data

Student	Homework Score (%)	Final Exam Score (%)
S1	80	75
S2	95	92
S3	70	78
S4	60	62
S5	85	82
S6	90	85
S7	75	80
S8	65	65
S9	55	68
S10	88	88



Find out Spearman's rank correlation coefficient.
What is the conclusion you can draw?

Problem 6



- ❖ An educational economist wants to establish the relationship between an individual's income and education. He takes a random sample of 10 individuals and asks for their income (in \$1000s) and education (in years). The results are shown below. Find the least squares regression line.

Education	11	12	11	15	8	10	11	12	17	11
Income	25	33	22	41	18	28	32	24	53	26



Thank you